

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	08 June 2026	
Team I D		
Project Name	Cosmetic Insights : Navigating Cosmetics Trends and Consumer Insights	
Max mum Marks	4 Marks	

Technical Architecture:

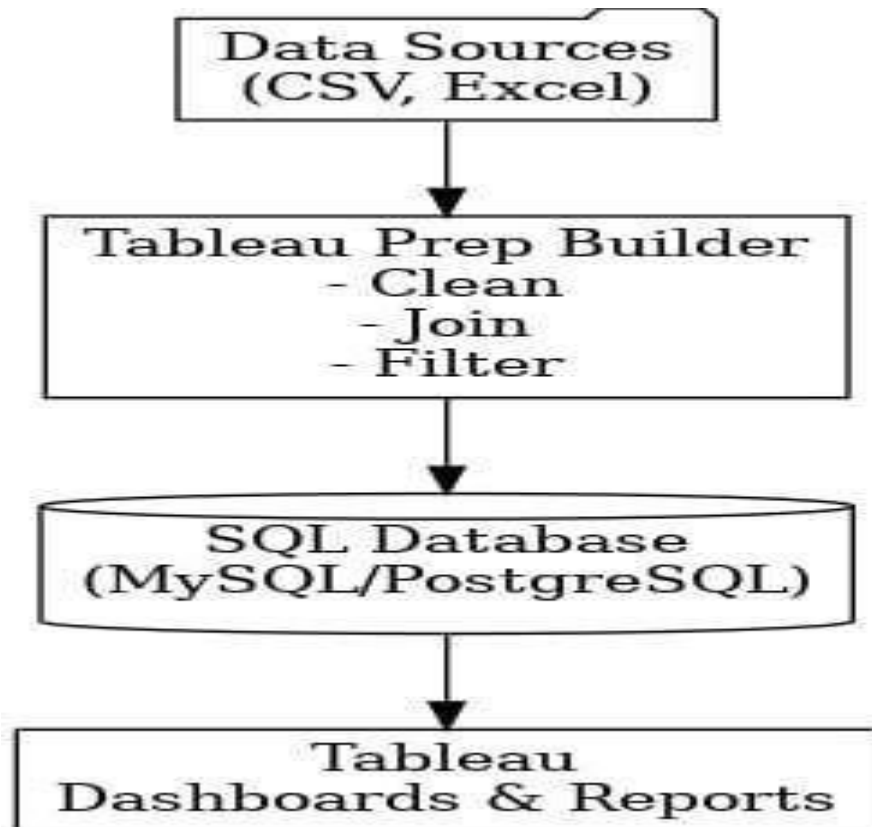


Table-1 : Components & Technologies:

S. No	Component	Description	Technology
1.	User Interface	Interactive dashboard where users analyze trends and filter by region, product	Tableau Embedded Dashboard, Bootstrap Web Portal
2.	Application Logic- 1	Automated data pipeline to pull sales data daily from CRM	online
3.	Application Logic-2	ETL and transformation: cleaning, aggregating, enriching data	Prep Builder
4.	Application Logic-3	Custom logic to calculate KPIs, YOY trends, and seasonal analytics	SQL Scripts / Tableau Calculated Fields
5.	Database	Stores sales, customer segmentation, pricing tier, and feedback data	PostgreSQL
6.	Cloud Database	Hosted database that syncs with Tableau	SQL , Excel
7.	File Storage	Raw datasets (CSV, Excel) and exported reports (PDF, PNG)	Drives
8.	External API- 1	Retrieves public skincare trend data from online sources	None
9.	External API-2	Integrates demographic or social target segmentation influence data for	None
10.	Machine Learning Model	Forecast product demand, predict user preferences	None
11.	Infrastructure (Server / Cloud)	Entire solution deployed as microservices with CI/CD pipelines	GitHub

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Tableau is a proprietary tool, so no open-source frameworks are used.	Not applicable
2.	Security Implementations	Tableau provides user authentication, row-level security , and data encryption for dashboards	Tableau User Permissions

S . No	Characteristics	Description	Technology
3.	Scalable Architecture	Tableau dashboards can connect to live data sources	Tableau Cloud / Tableau Server
4.	Ava il ab il ity	Tableau Cloud ensures high ava backups i lability , auto	Tableau Online
5.	Performance	Tableau optimizes performance via Hyper Extracts, Dashboard Performance	Tableau Hyper Engine,Extracts